

Agents, tools and retrieval, and where this course ends

A chatbot answers from what it absorbed in training. An agent can look things up, use tools, remember what happened, and check the result before it replies. Today you see each of those pieces as plain code you can read — and then we step back over thirteen weeks to what you can now build, and what you should be careful not to.

Unit 6 · Frontier

80 minutes

Demonstration · bridges into the agents labs

Your AI Tools lecture, part 2

Agent vs chatbot

Retrieval

Tools and loop

Limits and ethics

The agents labs

Before the final

PART 1 · 10 MINUTES

What makes an agent more than a chatbot

1



Tools

Functions it can call — a calculator, a database lookup, a search — for things that must not be guessed.



Memory

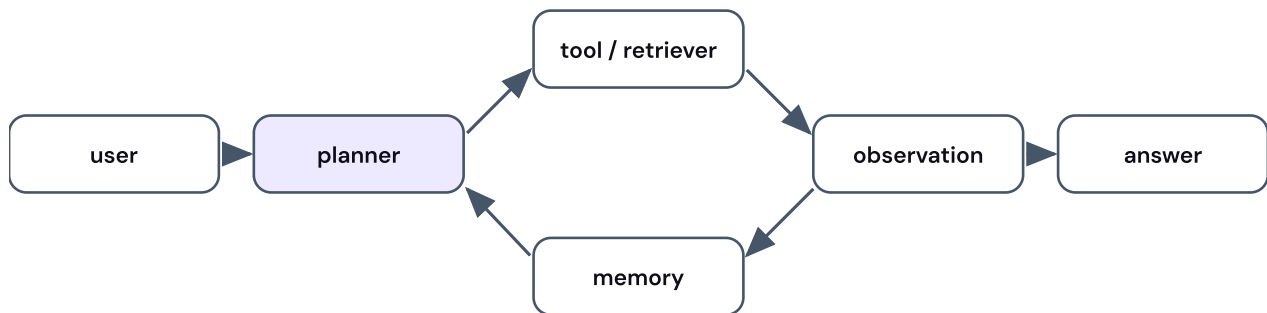
State that persists across steps: what was asked, what was tried, what came back.



A loop

Plan → act → observe → decide again, until the task is done or it should stop.

THE LOOP — PLAN, ACT, OBSERVE, REMEMBER, DECIDE AGAIN



each arrow is a place it can fail: wrong retrieval · wrong tool · bad argument · unsupported claim · stale memory · no stopping rule

PART 2 · 20 MINUTES · LIVE

Retrieval-augmented generation

"It does not know my documents."

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A model cannot know your course policies, your clinic's protocols or last week's memo. Retraining it for every document is impractical. **Retrieval-augmented generation** fetches the relevant passages at question time and puts them in the prompt, with instructions to use only them and cite them.

- 1 **Index** the documents so they can be searched — the demonstration uses TF-IDF so every score is inspectable; production systems use embeddings from session 24.
- 2 **Retrieve** the top passages for the question.
- 3 **Ground** the prompt: *use only the supplied context, cite a source label for every claim, say so if the context is insufficient.*
- 4 **Generate**, then check that each claim points at a source.

retrieval scores in the demonstration

TEXT

Can students use AI to write lab code?	→ ai_coding_policy.txt 0.52
Do students train a transformer from scratch?	→ transformer_scope.txt 0.388
What is the capital of Ghana?	→ transformer_scope.txt 0.0

RETRIEVAL ALWAYS RETURNS SOMETHING

The third query has nothing to do with the course, and the retriever still returned a document — with a score of zero. **An honest system acts on that score:** it says the documents do not cover the question, rather than letting a model write a confident answer around an irrelevant passage.

PART 3 · 15 MINUTES · LIVE

Tools, memory and an honest loop

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The demonstration agent — `CSC4315_S28_Demo_A_Small_Honest_Agent` — deliberately contains no language model. Its planner is three `if` statements, so every decision is a line you can read. It has a calculator that evaluates arithmetic and nothing else, a session lookup over the course map, retrieval, and a memory you can print.

three steps of the loop

TEXT

What happens in session 25?	→ lookup_session: Session 25 — Attention, computed by hand
calculate $17 * 240 / 100$	→ calculator: 40.8
Send an email to my professor	→ None: I have no tool that can do that, so I will not pretend to.

In a real agent a language model replaces the keyword planner, choosing tools and arguments from a description of each. That makes it far more flexible — and moves every failure on the diagram into a component you cannot read line by line. **Domain-specific agents** work best when the tools, documents and permissions are narrow and the checks are explicit.

Limits and ethics

Five questions to ask of any AI system you build or deploy.

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Question	Where the course met it
Would this feature exist at the moment of decision?	session 20 — leakage
Who bears the cost of each kind of mistake?	session 21 — precision, recall, thresholds
Does it work on the people and languages it will serve?	session 3 — the language gap; session 24 — tokenisers and bias
What happens to private data in training and prompts?	session 26 — fine-tuning corpora
Can a person check the output, and will they?	session 27 — verification as a habit

WHERE THIS COURSE ENDS

You arrived, many of you, unable to write a function. You have since built a neural network from matrix multiplications, derived and checked its gradients, trained it on real images, rebuilt it in three frameworks, broken and repaired it, evaluated it honestly, borrowed networks that see and write, computed attention by hand, and made a frontier model measurably better at a task. **None of it is magic to you any more — and that is the most useful thing an AI course can leave behind.**

The bridge: the agents component

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The six-lab agents curriculum runs outside class time and is the practical half of the final. It starts from today's notebook and replaces, one piece at a time, the keyword planner, the TF-IDF retriever and the toy tools with real ones — keeping the loop inspectable at every stage.

Before the final

1. **Session 29, Tuesday 15 December**, is held in reserve: a second pass on whatever needs it, or a supervised clinic on the agents component.
2. **Session 30, Friday 18 December, 11:30 — final examination**: a written paper plus the implementation component.
3. **Revise by thread**: the 2–3–1 network from session 8 to session 18, the honest–evaluation habits of unit 4, and the mechanism behind every language model — the next token.



Thank you

It has been a privilege to teach this course in Grand-Bassam. Build something that helps somebody — and check it.